Grounded Physical Reasoning for Robotic Manipulation with a Small Vision–Language Model

CIS 5300 Term Project Milestone 1

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Relative mass comparison is useful in robotic manipulation because it can affect grasping, moving, and stacking decisions. However, mass cannot be read directly from an image, and visual size may be misleading.Our project studies whether a small VLM, Qwen3-VL-2B-Instruct, can combine RGB images with the supplied physical dimensions and estimated thickness masks to decide which of two everyday objects is heavier. We ask three questions: whether the model should compare the two objects directly (pairwise) or first estimate each object's mass and then compare (pointwise), how much LoRA fine-tuning improves this judgment, and what the adapted model relies on, which we study through error analysis or probing. The following papers provide the dataset, the fine-tuning method, and the analysis tool for these questions.

image2mass: Estimating the Mass of an Object from Its Image

Standley et al. [1] estimate an object's mass from a single image, motivated by robots whose manipulation policies depend on the mass of the objects they handle. They collect about 3 million Amazon product listings, each with one image, a listed weight, and bounding-box dimensions, and filter them with ten heuristic stages and a learned classifier to about 150,000 products, removing problems such as listings that show multiple items or a product in its box. They also weigh and photograph 56 household objects as a second test set. Their shape-aware model multiplies a volume estimate by a density estimate: a geometry module predicts a per-pixel thickness map, which is combined with the listed dimensions in a volume tower, and an Xception-based density tower predicts density from the image. Training minimizes the absolute log difference between predicted and true mass. On the Amazon test set, the model predicts mass within a factor of two for 76.7% of items, compared with 48.8% for a nearest-neighbor baseline on pretrained Xception features. On the household objects it is competitive with individual human estimates (76.6% vs. 79.8% within a factor of two), while an ensemble of humans remains clearly better (87.7%).

This paper provides our dataset and the reference point for our pointwise setting, and its log-scale metrics, such as the fraction of predictions within a factor of two, give us a natural way to evaluate mass estimates. We retain the released images, physical dimensions, and estimated thickness masks, allowing the VLM to use both appearance and geometric information. Unlike the task-specific shape-aware network, our approach adapts a pretrained VLM with LoRA. The paper evaluates per-object estimates; in Milestone 2 we test whether a VLM that receives both objects and their associated geometry compares their masses more accurately than two independent estimates. We will also analyze errors involving material, contents, and the use of the supplied geometry.

LoRA: Low-Rank Adaptation of Large Language Models

Hu et al. [2] propose LoRA as an efficient alternative to full fine-tuning. Instead of updating every parameter, LoRA freezes a pretrained weight matrix W and learns a low-rank update $\Delta W = BA$. Only the much smaller matrices A and B are optimized, and their product can be merged into the original weights for inference. On GLUE, LoRA adapts RoBERTa-large with about 0.8 million trainable parameters and matches full fine-tuning of all 355 million parameters (average score 89.0 vs. 88.9). For GPT-3 175B, the authors report up to 10,000 times fewer trainable parameters and about three times lower training memory while maintaining competitive performance. Their ablations also give practical defaults: under a fixed parameter budget, adapting both the query and value projections (W_q and W_v) works best, and ranks as small as $r = 1$ suffice for GPT-3, although the best rank varies by task. Because only A and B need gradients and optimizer states, LoRA lets us fine-tune

a 2B-parameter VLM on a single 24 GB GPU, and we will tune r on the development set. The main limitation is that the paper evaluates text-only models, so our experiments must still establish whether low-rank updates help a small VLM combine visual features with supplied dimensions and thickness information for mass judgments.

Understanding Intermediate Layers Using Linear Classifier Probes

Alain and Bengio [3] introduce linear probes for analyzing intermediate neural-network representations. A linear classifier is trained on activations from each selected layer while its gradients are blocked from changing the original model. Probe accuracy measures how easily the target label can be decoded from a representation. In experiments on a small convolutional network, ResNet-50, and Inception-v3, class information generally becomes more linearly separable in deeper layers; probe error in ResNet-50 decreases almost monotonically across residual blocks. We will fit layer-wise linear regression probes for log mass on the zero-shot and LoRA-adapted VLMs, using the same image and geometry inputs, to examine where mass information becomes accessible as a complement to behavioral error analysis.

Data

We use the Amazon product data released with image2mass [1]. Our downloaded release contains 149,914 records (141,950 training, 6,529 validation, and 1,435 test). Each record stores a 512×512 RGB product image, the listed weight in pounds, the bounding-box dimensions in inches, an estimated 32×32 thickness mask, and a record ID. The products are varied consumer goods, including household and kitchen items, toys, tools, and electronics accessories such as pens, cups, bowls, stuffed toys, baking dishes, and cables. Weights range from 0.01 lb (4.5 g) to 14 lb (6.35 kg), with a median of 0.9 lb (0.41 kg); the middle 50% lie between 0.35 lb and 2.5 lb (0.16–1.13 kg). For example, record 101262 shows a wooden bowl with dimensions of 6 × 6 × 3 inches and a listed weight of 0.25 lb (0.11 kg).

We randomly sample 10,000 records within the original splits: 8,000 training records from the original training split, 1,000 development records from the original validation split, and 1,000 test records from the original test split (80/10/10). Each selected record retains its original split assignment. The training set is used to fit the model, the development set for parameter tuning and model selection, and the test set only for final evaluation. We retain the released records without additional image-similarity-based filtering or manual relabeling. Visually similar products with different dimensions and weights are retained, since the supplied geometry may distinguish different product sizes.

Models receive the decoded RGB image, physical dimensions, and supplied thickness mask during both fine-tuning and evaluation. Dimensions are provided as text with explicit units, and the thickness mask is encoded as an auxiliary visual input. The listed weight serves as the supervision target and evaluation ground truth; it is not provided as a prediction input. In the pointwise setting, the model estimates each object's mass independently before comparison. In the pairwise setting, it receives both objects' images, dimensions, and thickness masks and directly compares their masses, with comparison labels derived from their listed weights. Pairs are formed within each split. The file format and a full example record are documented in `data.md`.

Conclusion

Compared with [1], we replace a task-specific network with a pretrained VLM while retaining the released images, physical dimensions, and estimated thickness masks. We adapt the VLM with LoRA and add a pairwise setting in which the model compares two objects together with their associated geometry. Our main question is whether a small LoRA-adapted VLM compares mass more accurately by judging two objects directly or by estimating each first, and how it uses visual and geometric information.

References

[1] T. Standley et al. “image2mass: Estimating the Mass of an Object from Its Image.” *CoRL*, 2017.
[2] E. J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” *ICLR*, 2022.
[3] G. Alain and Y. Bengio. “Understanding Intermediate Layers Using Linear Classifier Probes.” *arXiv:1610.01644*, 2016.